GIDEON: Smart Offline Voice Assistant using Raspberry Pi

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Introduction

Gideon is a compact, offline voice assistant built on a Raspberry Pi that combines speech recognition, a locally hosted large language model (LLM), and real-time audio interaction to deliver intelligent, privacy-preserving responses. The system captures voice input through a microphone, processes it locally without internet dependency, and returns contextual answers both as on-screen text and spoken output via a Bluetooth speaker. By leveraging edge computing and efficient integration of hardware and AI, this project demonstrates a scalable, low-cost solution for real-time human–computer interactions in embedded systems. interaction in embedded systems.

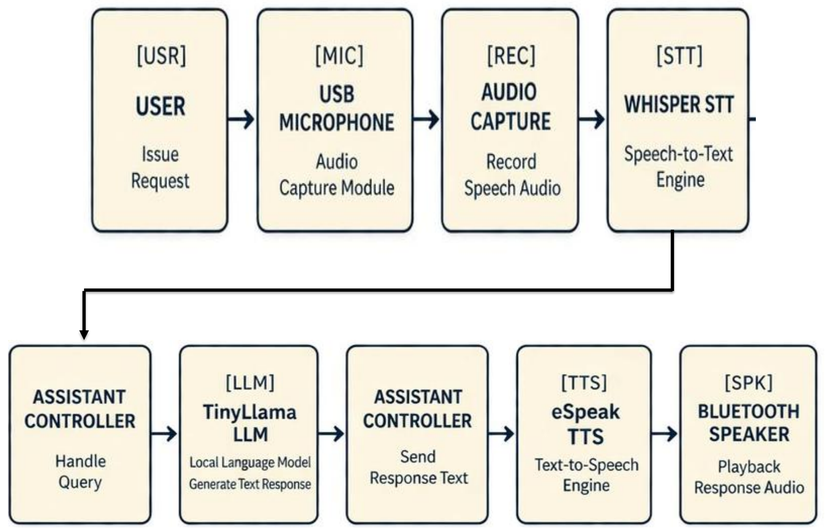
Objectives

- To develop an offline, voice-controlled AI assistant using Raspberry Pi.
- To implement real-time speech recognition and text-to-speech for natural interaction.
- To integrate a locally hosted large language model (LLM) for intelligent response generation without internet dependency.
- To design a low-cost, privacy-preserving system using edge computing for efficient human–computer interaction.

Methodology

- The Raspberry Pi is configured as the main processing unit with required libraries for audio processing, speech recognition, and local LLM execution.
- User voice input is captured through a webcam microphone and recorded using system-level audio tools.
- The recorded audio is processed using the Whisper model to convert spoken language into text accurately.
- The transcribed text is passed to a locally hosted TinyLlama large language model (LLM) for understanding and generating a context-aware response.
- The generated response is displayed on the monitor in real time for visual feedback.
- The response text is converted into speech using the eSpeak text-to-speech engine.
- The generated speech is transmitted through a Bluetooth speaker, enabling clear and natural user interaction.
- The entire system operates without internet connectivity, ensuring low latency, data privacy, and efficient edge-based processing.

Block Diagram/Figure



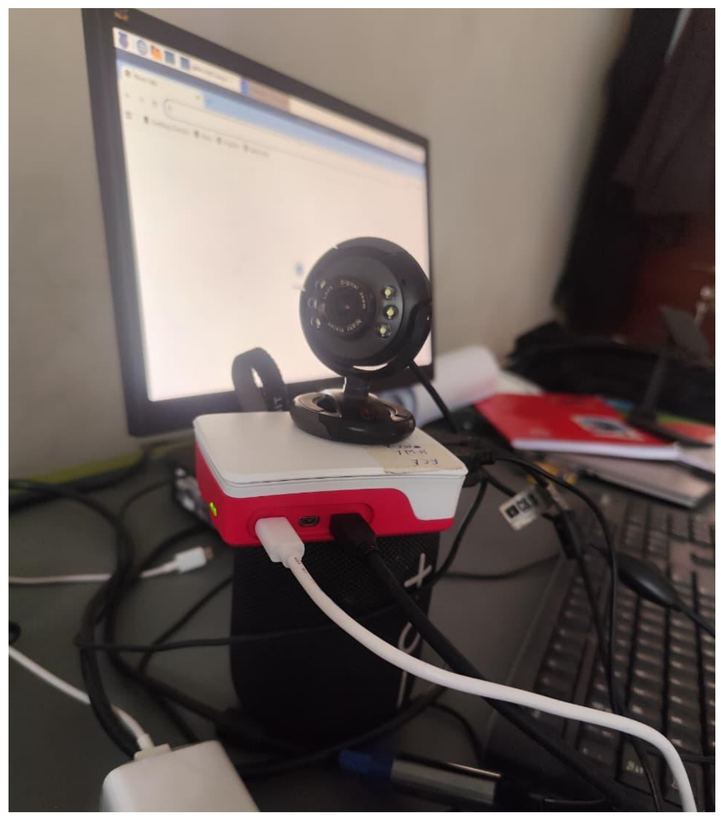


Figure. Developed hardware.

Results

1. Successful Voice Interaction
The system accurately captures user voice input and processes it into meaningful queries in real time.

2. Accurate Speech Recognition
The Whisper model demonstrates reliable speech-to-text conversion with good accuracy for clear spoken input.

3. Effective AI Response Generation
The locally hosted TinyLlama LLM generates relevant and context-aware responses without internet dependency.

4. Smooth Audio Output
The system successfully converts text responses into clear speech and delivers them through a Bluetooth speaker.

5. Fully Offline Operation
The assistant operates completely offline, ensuring user data privacy and independence from network connectivity.

6. Low-Cost and Compact Implementation
The entire system is implemented using a Raspberry Pi, proving feasibility of embedded AI at low cost.

7. Real-Time Performance
The system provides responses within a short time, demonstrating practical usability for everyday interaction.

Achievements

- Developed a fully offline, voice-controlled AI assistant.
- Integrated speech recognition, local LLM, and text-to-speech in one system.
- Achieved real-time voice interaction with both text and audio output.
- Implemented an efficient and low-cost edge AI solution using Raspberry Pi.
- Ensured user data privacy by eliminating dependency on cloud-based processing.

Conclusions

- The project successfully demonstrates the development of an offline, voice-controlled AI assistant using Raspberry Pi.
- Efficient integration of speech recognition, local LLM, and text-to-speech enables seamless human–computer interaction.
- The system operates entirely without internet, ensuring enhanced data privacy and reliability.
- Real-time response generation proves the practicality of deploying AI on edge devices.
- The implementation highlights a cost-effective approach to embedded AI systems.
- The project can be further extended for applications in smart homes, education, and assistive technologies.

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